LESSON 1

Oxidation and Reduction

ELECTRON TRANSFER, OXIDATION NUMBERS, AND REDOX AGENTS

Standards

Reactions: explain how electron transfer drives oxidation-reduction reactions.

Objectives

- What is the meaning of redox reactions?
- How do oxidation and reduction differ?
- What happens to the oxidation number of an atom when it is oxidized or reduced?
- What are oxidizing and reducing agents?

Expectations

- Define oxidation and reduction in terms of electron transfer.
- Track oxidation numbers to identify what is oxidized and what is reduced.
- Identify the oxidizing agent and the reducing agent in a redox reaction.
- Use electronegativity to determine which atom is oxidized and which is reduced in molecular redox reactions.

Section 1

Oxidation and Reduction

MAIN IDEA

A redox reaction is an electron transfer: the atom that loses electrons is oxidized, the one that gains them is reduced.

KWS

WHAT I KNOW	WHAT I WANT TO KNOW	SOURCE
		YouTube Crash Course Chemistry #10: Redox Reactions PhET PhET: Battery Voltage — electron transfer in action

Vocabulary

Review

- chemical reaction
- ion
- electronegativity
- valence electron

New

- oxidation
- reduction
- redox reaction
- oxidation number
- oxidizing agent
- reducing agent

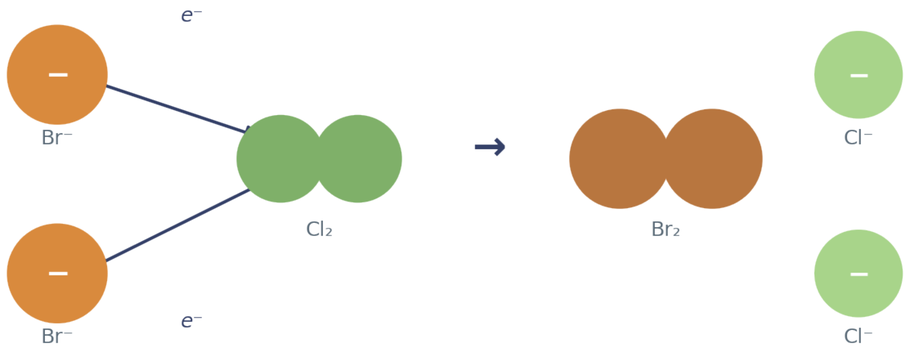
Group Investigation

- Form 3 groups. Each group researches the topic and shares its findings with the class.
- Each stage earns a point: research, discussion, solving the questions, and answering other groups' questions.
- Guiding questions: What is the meaning of redox reactions? How do oxidation and reduction differ? What happens to an atom's oxidation number when it is oxidized or reduced? What are oxidizing and reducing agents?

Electron Transfer and Redox Reactions

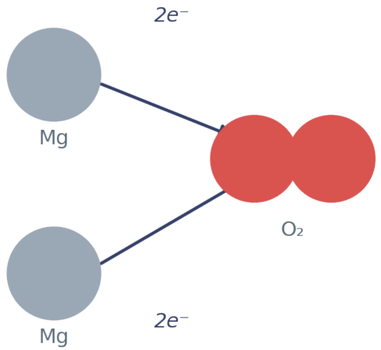
- An oxidation-reduction reaction, or redox reaction, involves the transfer of electrons from one atom to another.
- A reaction in which electrons are transferred from one substance to another is called an oxidation-reduction reaction, also called a redox reaction.

$2Br^- + Cl_2 \rightarrow Br_2 + 2Cl^-$



Br^- loses an electron (oxidized);
Cl_2 gains electrons (reduced)

$2Mg + O_2 \rightarrow 2MgO$



Each Mg loses 2 electrons (oxidized);
O_2 gains electrons (reduced)

Oxidation and Reduction

- Oxidation is defined as the loss of electrons from atoms of a substance.
- Reduction is defined as the gain of electrons by the atoms of a substance.
- Oxidation numbers are tools that scientists use to keep track of the movement of electrons in a redox reaction.
- The oxidation number of an atom in an ionic compound is the number of electrons lost or gained by the atom when it forms an ion.

Oxidation

$Na \rightarrow Na^{+} + e^{-}$

Reduction

$Cl2 + 2e^{-} \rightarrow 2Cl^{-}$

Tracking Oxidation Numbers

- Oxidation numbers rise and fall in exactly opposite directions for oxidation and reduction:

Process	What happens to the atom	Oxidation number
Oxidation	Loses electron(s)	Increases
Reduction	Gains electron(s)	Decreases

Identify Oxidation-Reduction Reactions

- The following equation represents the redox reaction of aluminum and iron:
- $2Al + 2Fe^{3+} + 3O^{2-} \rightarrow 2Fe + 2Al^{3+} + 3O^{2-}$
- Task: identify each of the following changes as either oxidation or reduction.

a	**$I2 + 2e^- \rightarrow 2I^-$**
b	$K \rightarrow K^+ + e^-$
c	$Fe^{2+} \rightarrow Fe^{3+} + e^-$
d	$Ag^+ + e^- \rightarrow Ag$

Oxidizing and Reducing Agents

- The substance that oxidizes another substance by accepting its electrons is called an oxidizing agent — it is itself reduced in the reaction.
- The substance that reduces another substance by losing its electrons is the reducing agent — it is itself oxidized in the reaction.

2K(s) + Cl2(g) → 2KCl(s)

Oxidizing agent: Cl2 (gets reduced) • Reducing agent: K (gets oxidized)

Table 1: Summary of Redox Reactions

Process	What Happens	Agent & Oxidation Number
Oxidation	X loses an electron.	X is the reducing agent and becomes oxidized; the oxidation number of X increases.
Reduction	Y gains an electron.	Y is the oxidizing agent and becomes reduced; the oxidation number of Y decreases.

Redox and Electronegativity

- Redox reactions are not limited to atoms of an element changing to ions — some redox reactions involve changes in molecular substances or polyatomic ions.
- To determine which atom was oxidized and which was reduced in a molecular reaction, you must know which atom is more electronegative.
- Elements with high electronegativity are strong oxidizing agents.

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

N is reduced (more electronegative); H is oxidized (less electronegative)

Checkpoint: Oxidation and Reduction

- 1. In your own words, explain why reduction involves a GAIN of electrons even though the number seems to “reduce.”
- 2. Identify the oxidizing agent and the reducing agent: $Fe(s) + 2Ag^{+}(aq) \rightarrow Fe^{2+}(aq) + 2Ag(s)$. Explain your answer.
- 3. For Mg + I2 → MgI2, identify which element is oxidized and which is reduced.

Quick check — discuss with a partner, then confirm with your teacher.

Lesson Summary

Key Takeaways

- A redox reaction transfers electrons from one atom or ion to another.
- Oxidation = loss of electrons (oxidation number increases). Reduction = gain of electrons (oxidation number decreases).
- The oxidizing agent is reduced; the reducing agent is oxidized.
- In molecular redox reactions, the more electronegative atom is reduced.

Vocabulary Recap

- oxidation — loss of electrons
- reduction — gain of electrons
- oxidation number — tracks electron loss/gain
- oxidizing agent — accepts electrons, is itself reduced
- reducing agent — gives up electrons, is itself oxidized

Evaluation: Oxidation and Reduction

Complete independently. Review the summary if you get stuck, then check your answers with your teacher.

- 1. Define oxidation and reduction in terms of electrons, and explain how the two always occur together.
- 2. Identify the oxidizing agent and the reducing agent: $H_2S + Cl_2 \rightarrow S + 2HCl$. Explain your reasoning.
- 3. For each half-reaction, state whether it represents oxidation or reduction: (a) $Fe \rightarrow Fe^{3+} + 3e^-$ (b) $O_2 + 4e^- \rightarrow 2O^{2-}$.
- 4. Explain why electronegativity is needed to identify oxidation and reduction in a molecular reaction like $N_2 + 3H_2 \rightarrow 2NH_3$, but not in an ionic reaction like $2Na + Cl_2 \rightarrow 2NaCl$.
- 5. Challenge: In the reaction $2Al + 2Fe^{3+} + 3O^{2-} \rightarrow 2Fe + 2Al^{3+} + 3O^{2-}$, identify what is oxidized, what is reduced, the oxidizing agent, and the reducing agent.

Self-Evaluation Rubric

4 – Confident mastery • 3 – Solid grasp, minor gaps • 2 – Needs review • 1 – Revisit the lesson

oxi numbers

losing e^- (negative charge) ↑

Gain e^- (positive charge) ↓

Identify Oxidizing agent and reducing agent

$$\overset{0}{Fe}[s] + 2\overset{1+}{Ag}^+[aq] \longrightarrow \overset{2+}{Fe}^{2+} + 2\overset{0}{Ag}[s]$$

Fe $\underline{0} \longrightarrow \underline{2+}$

Oxidation — Reducing agent

lose e^-

$Ag^{+}{}_{1+} \longrightarrow 0$

Reduction — oxidizing agent

Gain

ionic

Metal + Non-Metal

1	2	13
Li	Be	Al
Na	Mg	
K	Ca	
+1	2+	3+

حفظ

Covalent

Non-Metals

Sharing

EN

حفظ

N^{3-} O^{2-} F^{-1}

P Cl

Br

I

1-

(H_2O_2)

-1

-3 3+ +1

NH_3

NO_2 | NO_3

$4+(2-x2)=0$ | $6+(2-x3)=0$

SO_3^{2-} | SO_4^{2-}

$4+(2-x3)=2-$ | $6+(2-x4)=2-$

Identify the oxidizing and reducing agent

$\overset{0}{Mg} + \overset{0}{I_2} \rightarrow \overset{2+2-}{MgI_2}$

Mg

Simple Explanation

Losing electron → oxidation number go up

$Fe \rightarrow Fe^{2+}$ [oxidation] [Reducing agent]

Gaining electron → oxidation number go down

$Ag^{+} \rightarrow Ag^{0}$ [Reduction] [oxidizing agent]

$Fe^{0}(s) + 2Ag^{1+}(aq) \rightarrow Fe^{2+} + 2Ag(s)$

$Fe^{0} \rightarrow 2+$

oxidization

Reducing

$2Ag^{+} \rightarrow 2Ag^{0}$

$1+ \rightarrow 0$

Reduction

oxidizing